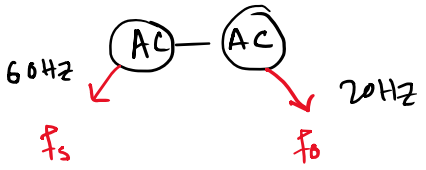
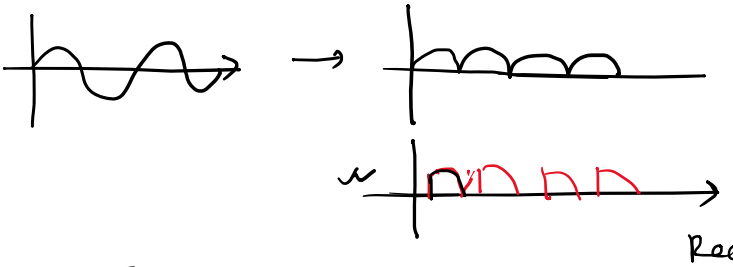
Cycloconverters

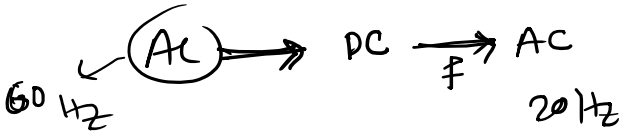
Friday, May 1, 2026 7:16 PM



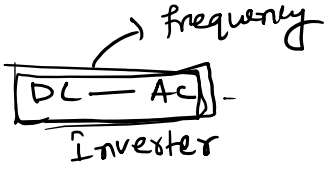
AC – AC

1) V_o (rms)
2) frequency

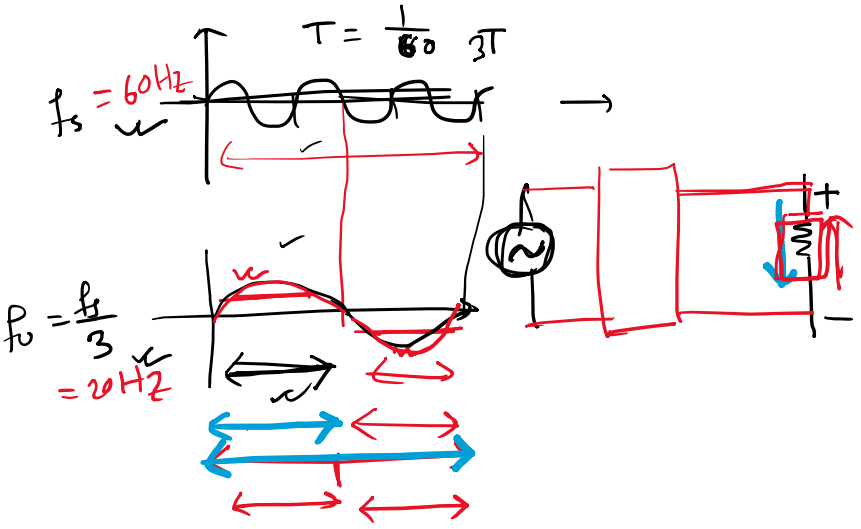








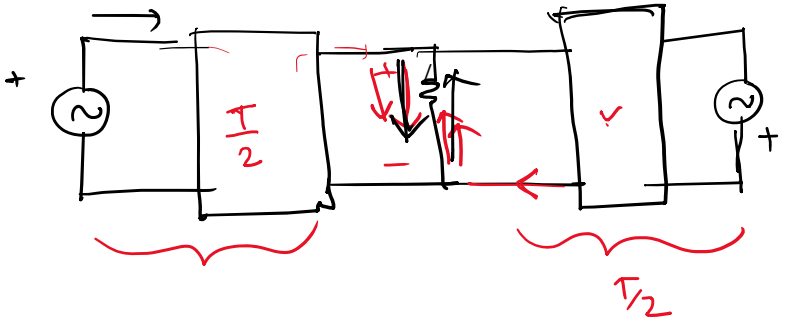
DC-Link



$f_o = \frac{f_s}{3}$

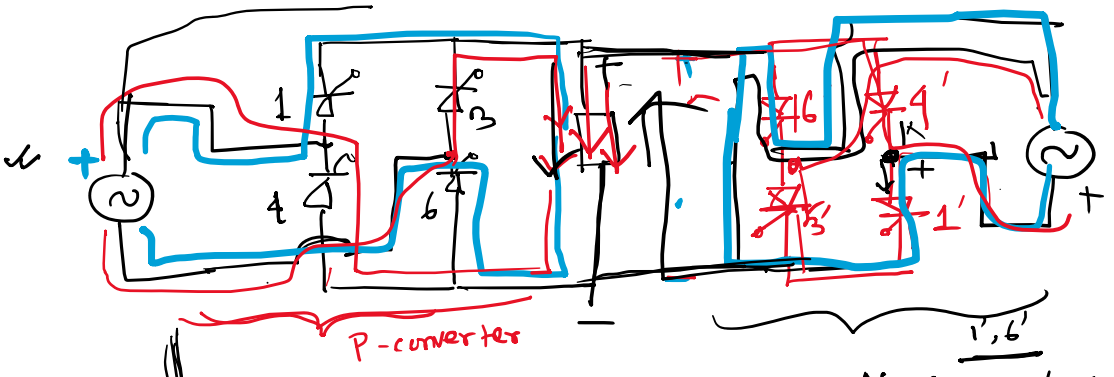
$T_o = 3T_s$

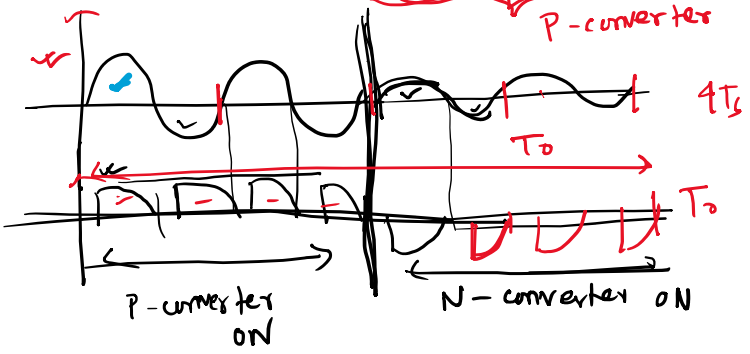
#



#

1,6





$T_0 = 4T_s$

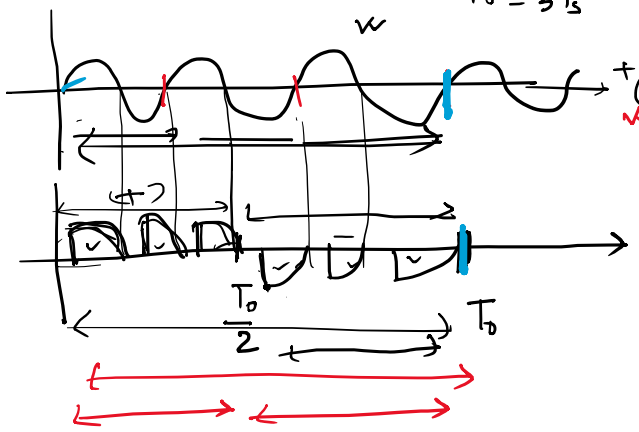
$\Rightarrow f_0 = \frac{f_s}{4}$

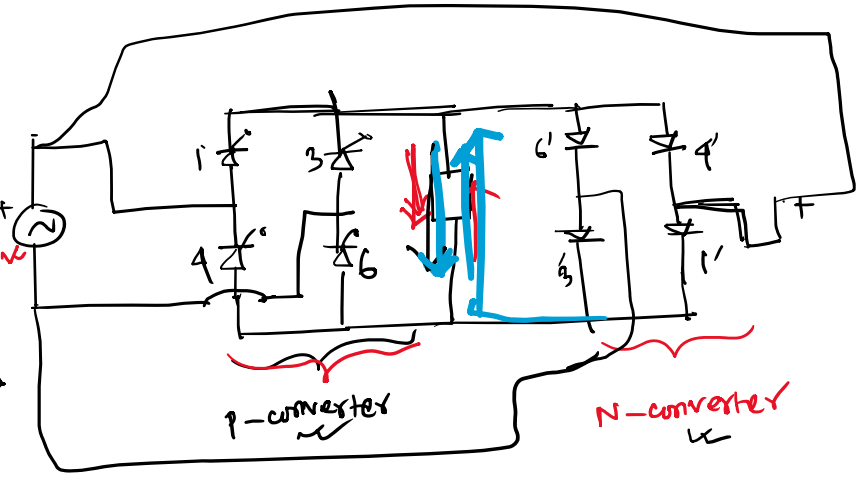
##

$f_0 = \frac{f_s}{3}$

$T_0 = 3T_s$

3T





$T_0 = 3T_s$

$\Rightarrow f_0 = \frac{f_s}{3}$

$0 - \frac{T}{2}$: ↓

$\frac{T}{2} - T$: ↑

$f_0 = \frac{f_s}{5}$

$T_0 = 5T_s$

3 phase cycloconverter

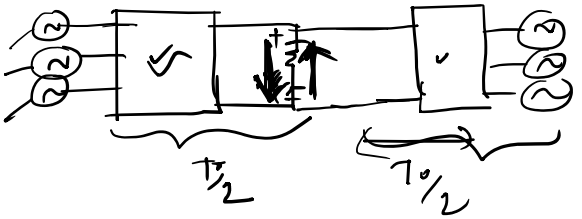
3φ – 1φ

60Hz

↓

30Hz

T_0

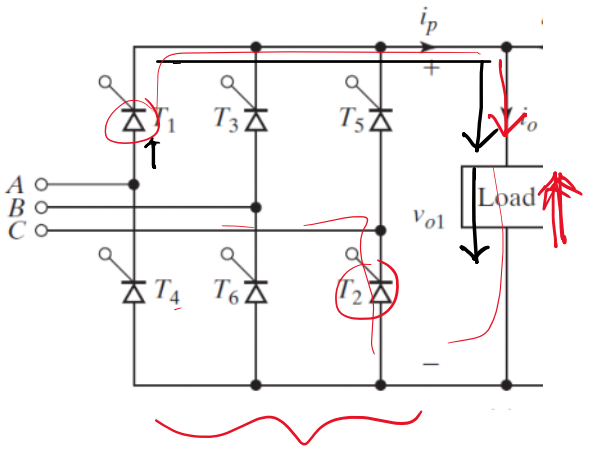


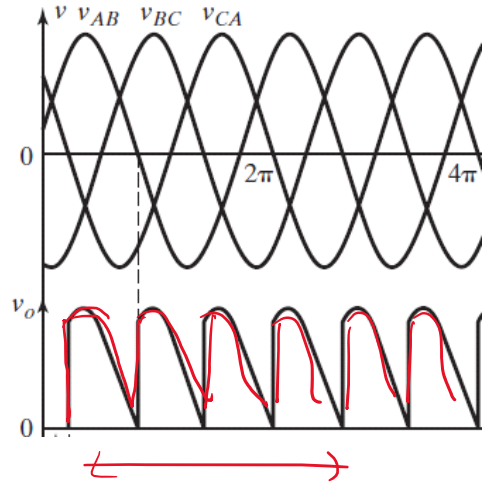
##





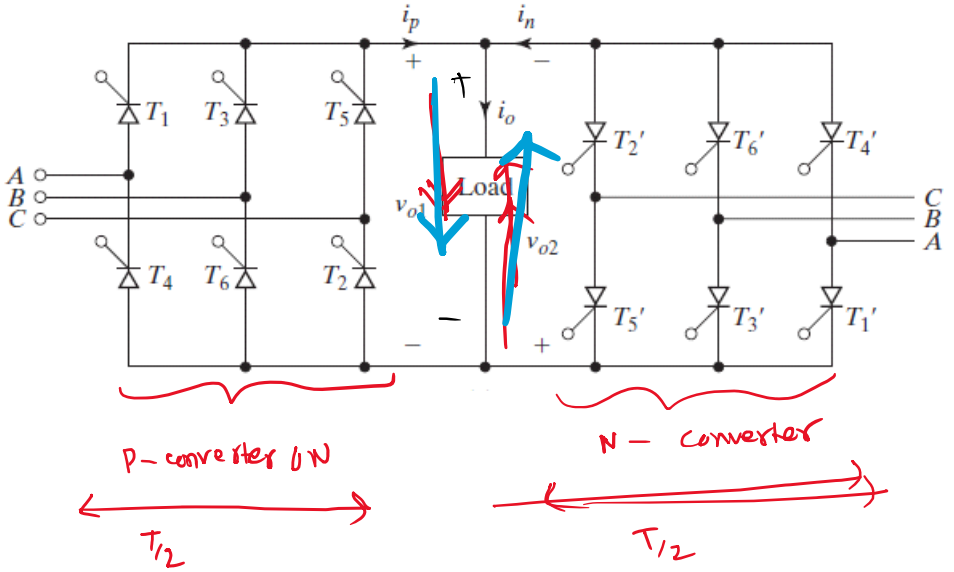
##





#

60Hz



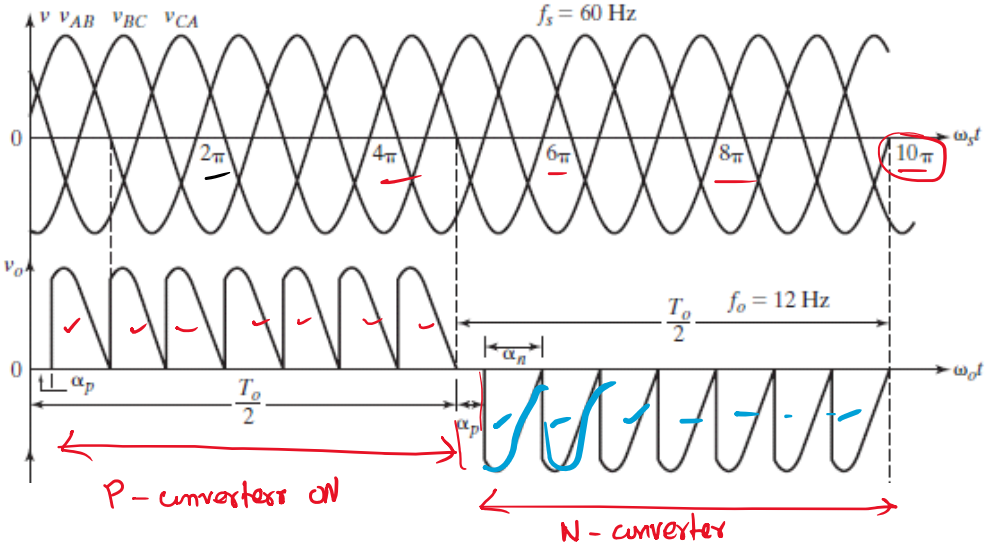
12Hz

$f_o = \frac{f_s}{5}$

$T_o = 5T_s$

3φ ←

1−φ ←



$T_o = 5T_s$